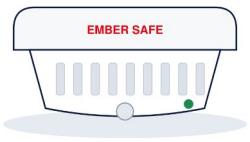
DATASHEET | GUARANTEED TECHNICAL PARTICULARS

SmartLoop Devices

Analogue addressable field devices for the SmartLoop-4 panel

All SmartLoop loop devices carry a **32-bit serial number** on their barcode label and take their address (001 to 128) from the panel - by auto-learn or by entering the serial number. No DIP switches and no hand-held programmer. Detectors, call points and modules have a **built-in short-circuit isolator**.



ES-MSD100 Heat & Smoke Multi-Sensor

Analogue addressable multi-sensor for relay rooms, power equipment rooms, battery rooms, ASM rooms, DG rooms, CCTV and OFC equipment rooms.

ES-MSD100 - TECHNICAL DATA	
Smoke sensing	Photo-electric (optical) chamber; response within 1.6 to 12.5 % obs/m (0.5 to 4.0 % obs/ft)
Sensitivity	4 levels set from the panel: High 1.5, Normal 2.5 (default), Low 3.5, Very low 4.0 % obs/ft
Directional response	Ratio of highest to lowest response threshold value 1.6 or less
Heat sensing	Fixed temperature 58 deg C and rate-of-rise 6 to 11.1 deg C/min, independent of the smoke sensing
False alarm rejection	Every alarm checked again before it is sent; automatic compensation for slow dust build-up; dirty-chamber warning
Pre-alarm	Pre-alarm report at 50 % of the alarm level
Indicator	Green blink = healthy; red steady = alarm (visible from 6 m, stays on until reset). A failed indicator does not stop the fire signal
Isolator	Built-in short-circuit isolator
Address	001 to 128; set from the panel (auto-learn or barcode serial number)
Supply	Loop powered, 24 V DC nominal
Insulation resistance	10 M ohm or more
Housing	Flame-retardant plastic; does not soften below 95 deg C (IS 2175 cl. 4.2); white
Mounting	Surface mounting base for ceiling or wall, held independent of the wiring
Operating conditions	0 to 49 deg C; up to 85 % RH
Size	Diameter _____ mm, height _____ mm (with base)

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ES-MCP100 Manual Call Point

Addressable manual call point for entrances, exits and escape routes.

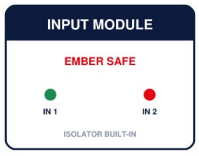
ES-MCP100 - TECHNICAL DATA	
Operation	Push the resettable element; reset with the supplied key - no glass to replace
Indicator	Green blink = healthy; red steady = activated
Isolator	Built-in short-circuit isolator
Address	001 to 128; set from the panel
Action	Starts all hooters and strobes (action level 3)
Mounting	Surface back box; 1400 mm +/- 200 mm from floor to the middle of the call point
Housing	Red flame-retardant plastic
Operating conditions	0 to 49 deg C; up to 85 % RH
Size	_____ x _____ x _____ mm



ES-HS100 Hooter cum Strobe

Combined audio and visual alarm for the Station Master room, relay room and ESM duty room.

ES-HS100 - TECHNICAL DATA	
Hooter	Piezo-electric; sound level adjustable up to 90 dB at 1 m
Strobe	Flashing red / white
Connection	Panel monitored sounder circuit, 24 V DC; end-of-line supervision
Operating conditions	0 to 49 deg C; up to 85 % RH
Size	_____ x _____ x _____ mm

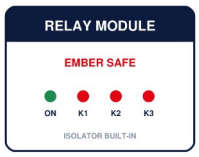


ES-IM100 Input Module

Brings other fire detection equipment onto the SmartLoop-4 loop as addressable points.

ES-IM100 - TECHNICAL DATA	
Inputs	2 monitored inputs (open / short / active), each with its own address
Use	LHS cable or LHD interface, aspirating smoke detector stages, UV / IR flame detectors, probe-type bi-metallic heat detectors, suppression system feedback
Isolator	Built-in short-circuit isolator
Indicator	Green blink = healthy; red = input active
Operating conditions	0 to 49 deg C; up to 85 % RH

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ES-RM100 Relay Output Module

Addressable outputs for automatic fire suppression, power shut-down and the data logger.

ES-RM100 - TECHNICAL DATA	
Contacts	3 NO + 3 NC potential-free contacts, at least 500 mA each
Control	Programmable from the panel: action level, delay (0 to 10 min in 0.5 min steps), room
Isolator	Built-in short-circuit isolator
Indicator	Green blink = healthy; red = output operated
Operating conditions	0 to 49 deg C; up to 85 % RH



ES-GSM100 GSM Module (optional)

SMS alarm messages to railway staff.

ES-GSM100 - TECHNICAL DATA	
Network	Quad-band GSM 850 / 900 / 1800 / 1900 MHz; SIM provided by the purchaser
Messages	SMS to 5 or more programmable mobile numbers for action levels 2, 3 and 4, within 30 s; repeated at once if sending fails
Supply	From the panel
Operating conditions	0 to 49 deg C; up to 85 % RH

Ordering information

Code	Product	Description
ES-MSD100	Heat & Smoke Multi-Sensor	Analogue addressable optical smoke + fixed / rate-of-rise heat detector with built-in isolator
ES-MCP100	Manual Call Point	Addressable, resettable manual call point with built-in isolator
ES-HS100	Hooter cum Strobe	Piezo-electric hooter, up to 90 dB at 1 m, with red / white flashing strobe
ES-IM100	Input Module	Addressable monitored input module for LHS / LHD interfaces, aspirating detectors, flame and probe heat detectors, suppression feedback
ES-RM100	Relay Output Module	Addressable relay module, 3 NO + 3 NC potential-free contacts, for suppression release, power shut-down and data logger
ES-GSM100	GSM Module (optional)	Quad-band GSM modem: SMS alarms to 5 or more mobile numbers
ES-SLPC	SmartLoop PC Software	Windows software: site layout view, event download and analysis, configuration
ES-BAT	Standby Battery	2 x 12 V sealed maintenance-free lead-acid batteries, sized for 24 h standby + 15 min alarm

Marking on each device: manufacturer name and address, year of manufacture, serial number (with barcode) and RDSO/SPN/217/2025 Version 3.1. Warranty 24 months from the date of supply.

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INFUSIVE INDOTECH PRIVATE LIMITED **Ember Safe | Stay Alert**

Address: _____ Phone: _____

E-mail: _____ Website: _____ Designed and manufactured in India. Data may change without notice.

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